



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Case Study No.: 6 Name: Dhanish Ladwani PRN: 24070521020

1. Experiment Title:

Develop web applications using JavaScript to reverse a string and count the number of vowels in a paragraph using string methods and regular expressions.

2. Software/Tools Required:

Google Chrome (Web Browser), Visual Studio Code (Code Editor), Browser Developer Tools / Console, HTML, CSS, Javascript

3. Experiment Program Code:

Case Study 6.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Experiment 6 - Case Studies</title>
  <link rel="stylesheet" href="Case Study 6.css">
</head>
<body>
  <div class="container">
    <h2>Experiment 6 - Case Studies</h2>

    <div class="box">
      <h3>Case Study 1: Reverse a String</h3>
      <label for="strInput"><b>Enter a String:</b></label><br>
      <input type="text" id="strInput" placeholder="Enter string to reverse">
      <br>
      <button onclick="reverseString()">Reverse String</button>
      <div id="result"></div>
    </div>

    <div class="box">
      <h3>Case Study 2: Vowel Counter in Paragraph</h3>
      <label for="paragraphInput"><b>Enter a Paragraph:</b></label><br>
      <textarea id="paragraphInput" rows="4" placeholder="Enter a paragraph to count
vowels..."></textarea>
      <br>
      <button onclick="countVowels()">Count Vowels</button>
      <div id="vowelResult"></div>
    </div>

    <footer style="text-align: center; margin-top: 30px; font-size: 14px; color: gray; border-
top: 1px solid lightgray; padding-top: 15px;">
      Made by Dhanish Ladwani PRN - 24070521020
    </footer>
  </div>

  <script src="Case Study 6.js"></script>
</body>
</html>
```

Case Study 6.css

```
body {
  font-family: Arial, sans-serif;
  margin: 40px;
  background-color: whitesmoke;
}

h2 {
  text-align: center;
  color: darkblue;
}

h3 {
  color: darkblue;
  margin-top: 0;
  margin-bottom: 15px;
  border-bottom: 1px solid #eee;
  padding-bottom: 5px;
}

.container {
  max-width: 650px;
  margin: 0 auto;
  background: white;
  padding: 20px;
  border-radius: 8px;
  border: 1px solid lightgray;
  box-shadow: 0 4px 10px rgba(0, 0, 0, 0.1);
}

.box {
  border: 1px solid #ccc;
  padding: 20px;
  margin-top: 20px;
  border-radius: 8px;
  background-color: #fafafa;
}

input, textarea, button {
  padding: 10px;
  margin: 10px 0;
  width: 95%;
  font-family: inherit;
  box-sizing: border-box;
}

textarea {
  resize: vertical;
}

button {
  width: 150px;
  background-color: royalblue;
  color: white;
  border: none;
  border-radius: 4px;
  cursor: pointer;
  font-weight: bold;
}

button:hover {
  background-color: blue;
}

#result, #vowelResult {
  margin-top: 15px;
  padding: 15px;
  background-color: #ffffff;
  border: 1px solid #ddd;
}
```

```
border-radius: 6px;
font-size: 15px;
}
```

Case Study 6.js

```
function reverseString() {
    let str = document.getElementById("strInput").value;

    if (str.trim() === "") {
        document.getElementById("result").innerHTML = "<b style='color:red;'>Please enter a string!</b>";
        return;
    }

    let reversedStr = str.split("").reverse().join("");

    document.getElementById("result").innerHTML = `
        <p><b>Original String:</b> ${str}</p>
        <p><b>Reversed String:</b> ${reversedStr}</p>
    `;
}

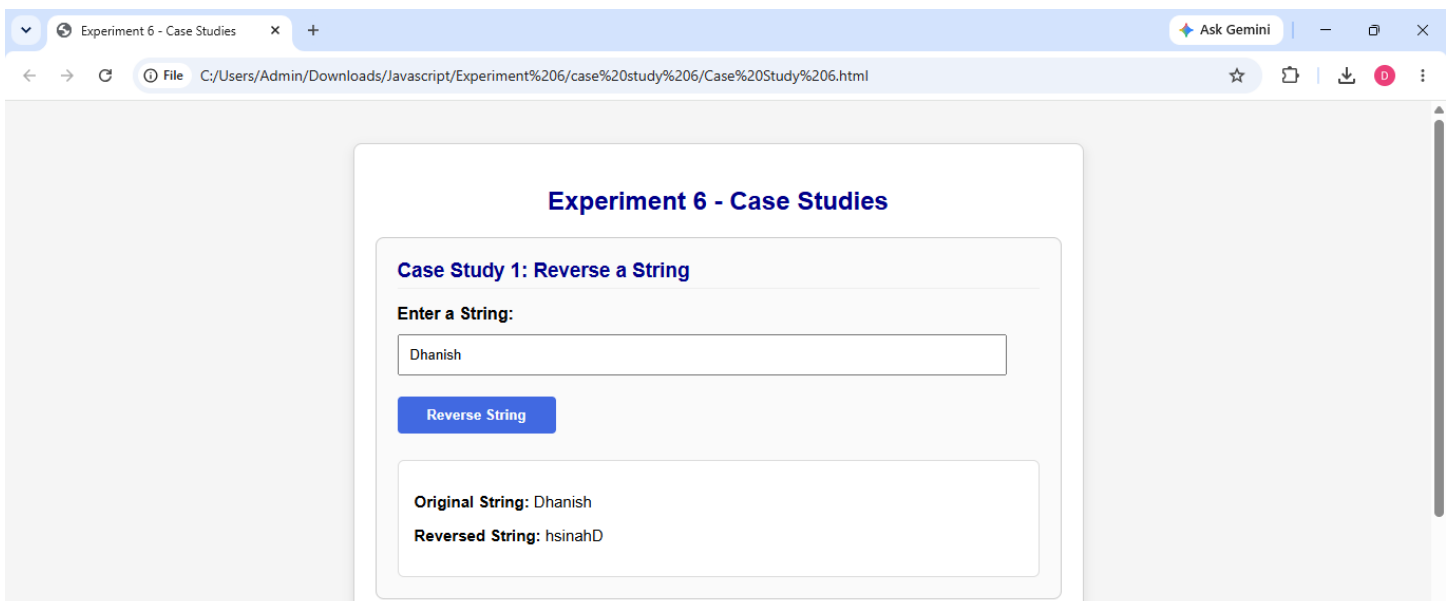
function countVowels() {
    let paragraph = document.getElementById("paragraphInput").value;

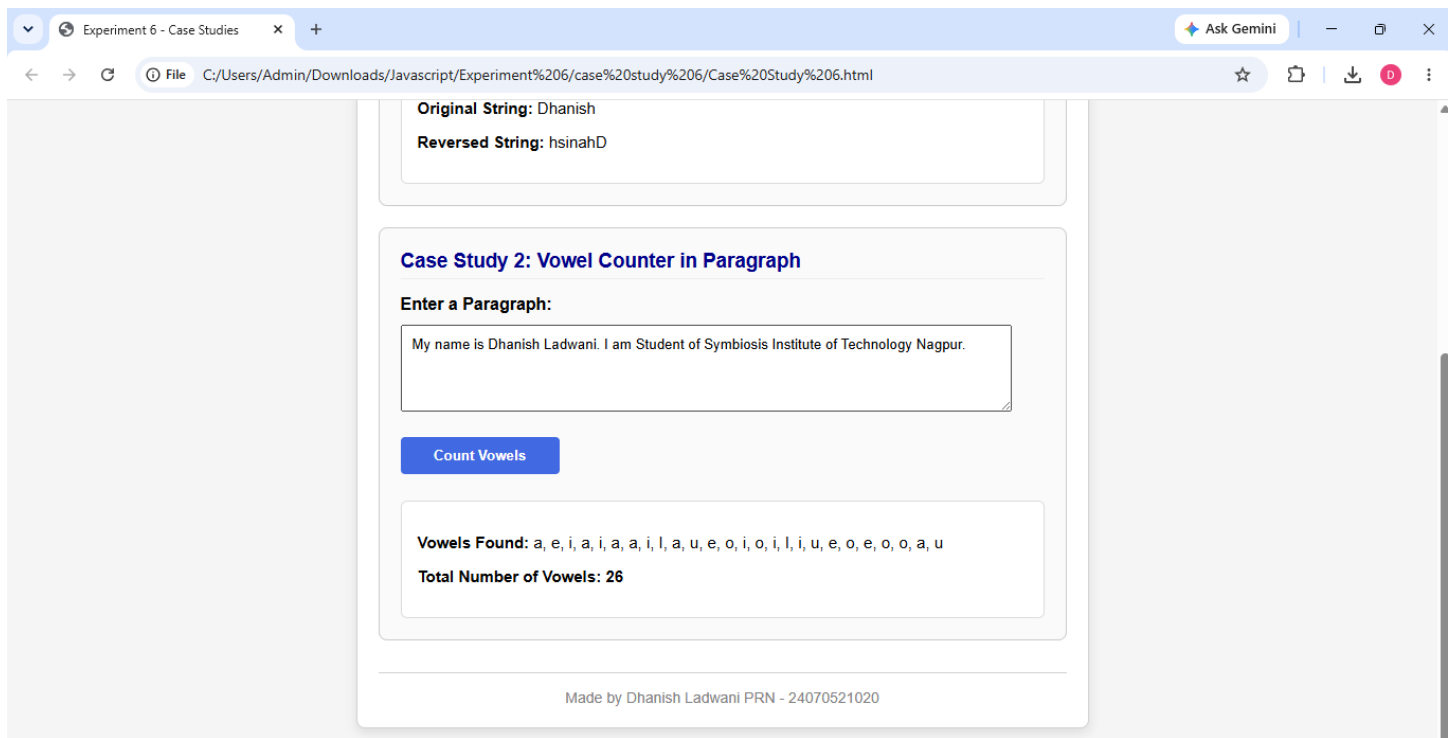
    if (paragraph.trim() === "") {
        document.getElementById("vowelResult").innerHTML = "<b style='color:red;'>Please enter a paragraph!</b>";
        return;
    }

    let vowels = paragraph.match(/[aeiou]/gi);
    let vowelCount = vowels ? vowels.length : 0;
    let vowelsList = vowels ? vowels.join(", ") : "No vowels found";

    document.getElementById("vowelResult").innerHTML = `
        <p><b>Vowels Found:</b> ${vowelsList}</p>
        <p><b>Total Number of Vowels:</b> <b>${vowelCount}</b></p>
    `;
}
```

4. Output:





5. Result/Conclusion:

Two small case studies were implemented to reinforce string handling in JavaScript. The first uses the `split("")`, `reverse()` and `join("")` chain to reverse a user-entered string, with a validation check for empty input. The second uses the `match()` method with a regular expression to find all vowels in a paragraph, displaying the matched vowels and their total count. Both case studies reinforced core string methods and simple regular expression pattern matching for text processing in JavaScript.